PROJECT DOCUMENTATION • PARTS 1–3

AI FITNESS FORM CORRECTOR

An AI-powered Computer Vision and Intelligent Fitness Assistant

Project Planning & Management
Literature Review
Requirements Gathering

Contents

1. Project Planning & Management 4
1.1 Project Proposal 4
1.2 Project Objectives 4
1.3 Project Scope 4
1.4 Project Plan and Timeline 5
1.5 Milestones and Deliverables 7
1.6 Resource Allocation 8
1.7 Task Assignment & Roles 8
1.7.1 Mahmoud 8
1.7.2 Mostafa 9
1.7.3 Mohamed 9
1.7.4 Hashem 10
1.7.5 Islam 10
1.8 Risk Assessment & Mitigation Plan 10
1.9 Key Performance Indicators (KPIs) 11
2. Literature Review 12
2.1 Review Approach 12
2.2 Similar Solutions and Projects 12
2.3 Comparative Analysis 13
2.4 Lecturer Feedback & Evaluation 13
2.5 Suggested Improvements 14
2.6 Proposed Final Grading Criteria 14
3. Requirements Gathering 15
3.1 Stakeholder Analysis 15
3.2 User Stories 15
3.3 Use Cases 16
3.4 Functional Requirements 16
3.5 Non-functional Requirements 17
3.6 Assumptions and Constraints 17
Requirements Traceability Matrix 18
References 19

1. Project Planning & Management

1.1 Project Proposal

AI Fitness Form Corrector is an intelligent Computer Vision system that analyzes a user's exercise movements through a camera and provides real-time feedback on exercise form. The system detects human body landmarks using pose estimation, transforms these landmarks into movement features such as joint angles and relative positions, recognizes the exercise being performed, classifies the form as correct or incorrect, counts repetitions, and provides corrective feedback.

The minimum viable product (MVP) focuses on Squat, Push-up, and Biceps Curl. The planned pipeline is: Camera/Video → Privacy Layer → Pose Estimation → Body Keypoints → Feature Engineering → Exercise Recognition → Form Classification → Rep Counting → Feedback → User Interface.

> **Project Value**
> The project combines Computer Vision, Machine Learning, real-time processing, MLOps, cloud deployment, privacy-aware processing, and an NLP/LLM assistant into one end-to-end system, delivered by a five-member team in a seven-week timeline.

1.2 Project Objectives

- Detect and track a user's body pose in real time from a camera feed.
- Recognize Squat, Push-up, and Biceps Curl.
- Extract joint angles, relative positions, movement range, and alignment features.
- Classify exercise form as correct or incorrect and identify major form issues.
- Count repetitions using pose-based movement states and angle thresholds.
- Provide immediate and understandable corrective feedback.
- Add privacy-aware processing through face/body/background blurring where appropriate.
- Provide an NLP/LLM assistant that can explain detected mistakes and answer fitness-related questions in context.
- Track experiments and model versions using MLflow and package the system using Docker.
- Deploy the integrated application to Microsoft Azure.
- Deliver the complete, documented, and demonstrable system by the deadline of **December 7, 2026**.

1.3 Project Scope

In Scope — MVP (Must Have)

- Three exercises: Squat, Push-up, Biceps Curl.
- Single-person pose tracking.
- Good/Bad form classification.
- Joint-angle and movement-feature extraction.
- Rule/state-machine-based repetition counting.
- Real-time feedback and a simple user interface.

- Subject-independent train/validation/test split where possible.
- Evaluation using Accuracy, Precision, Recall, F1-score, Confusion Matrix, and repetition-count error.
- MLflow experiment tracking, Git/GitHub version control, Docker packaging, and Azure deployment.

In Scope — Integrated Extensions (Should Have)

Because of the seven-week timeline, the following components are delivered in a focused, simplified form after the core MVP is stable:

- Privacy mode with face/body/background blurring.
- NLP/LLM assistant that explains detected mistakes using structured exercise-analysis context.

Advanced / Future Scope

- More exercises and additional incorrect-form categories.
- Temporal deep learning using LSTM, GRU, or Transformer models.
- Personalized workout and progress tracking.
- Voice interaction and voice feedback.
- Mobile application and edge/on-device inference.
- Enhanced privacy controls and broader anonymization.
- Advanced conversational coaching with long-term user memory.

1.4 Project Plan and Timeline

The project runs for **7 weeks**, from **Monday, October 19, 2026** to **Sunday, December 6, 2026**. The final submission deadline is **Monday, December 7, 2026**, which is 50 days after the date of this plan (October 18, 2026).

Milestones are defined as project deliverables, not learning stages. Seven overlapping milestones (M1–M7) are scheduled so that one milestone is completed at the end of every week. The first two milestones focus on planning, requirements, and data before the pose and AI baseline.

Schedule Principles

- **Parallel execution:** all five members contribute to every milestone through role-specific tracks (see Section 1.7). No milestone is owned by a single person.
- **Interface-first development:** data formats and API contracts are agreed in Week 1 so that every track can progress without waiting for another module.
- **Early setup:** the Git/GitHub workflow, project environment, and MLflow structure are set up in M2, and the MVP is integrated end-to-end in M4, leaving Weeks 5–7 for advanced features, deployment, validation, and finalization.
- **Feature freeze:** all new features stop at the end of Week 6 (November 29). Week 7 is reserved for testing, bug fixing, documentation, and demo rehearsal.

Sprint Calendar

Week	Dates	Active Milestones	Gate Review (End of Week)
W1	Oct 19 – Oct 25	M1 Planning & Requirements; M2 starts	M1 complete
W2	Oct 26 – Nov 1	M2 Data & Preprocessing; M3 starts	M2 complete
W3	Nov 2 – Nov 8	M3 Pose & AI Baseline; M4 starts	M3 complete
W4	Nov 9 – Nov 15	M4 MVP Integration; M5 starts	M4 complete
W5	Nov 16 – Nov 22	M5 Advanced Features; M6 starts; testing & documentation start	M5 complete
W6	Nov 23 – Nov 29	M6 Deployment & Validation; M7 starts	M6 complete; feature freeze
W7	Nov 30 – Dec 6	M7 Finalization; final testing, documentation, rehearsal	M7 complete

Gantt Chart

Work Package	Window	W1	W2	W3	W4	W5	W6	W7
M1 — Planning & Requirements	W1	■						
M2 — Data & Preprocessing	W1–W2	■	■					
M3 — Pose & AI Baseline	W2–W3		■	■				
M4 — MVP Integration	W3–W4			■	■			
M5 — Advanced Features	W4–W5				■	■		
M6 — Deployment & Validation	W5–W6					■	■	
M7 — Finalization	W6–W7						■	■
Testing, Documentation & Presentation	W5–W7					■	■	■

*Final submission deadline: **Monday, December 7, 2026**. Shaded cells show the weeks in which each work package is active.*

1.5 Milestones and Deliverables

Every milestone is delivered jointly by all five team members. The role-specific tasks behind each milestone are listed in Section 1.7.

Milestone	Window & Target	Deliverables	Acceptance Criteria
M1 Planning & Requirements	Oct 19 – Oct 25 Target: End W1	Requirements, system architecture, AI pipeline definition, scope, success criteria, CV/ML/backend/UI requirements.	Baseline approved by the team.
M2 Data & Preprocessing	Oct 19 – Nov 1 Target: End W2	Dataset analysis, label/class definition, preprocessing pipeline, feature schema, train/validation/test split, Git/GitHub workflow, MLflow structure.	Reproducible data pipeline; environment runs on every member's machine.
M3 Pose & AI Baseline	Oct 26 – Nov 8 Target: End W3	Validated pose keypoints (MediaPipe/YOLO), baseline models (Random Forest, SVM, XGBoost), initial real-time pipeline, initial UI.	Stable keypoint extraction; baseline evaluated on held-out subjects.
M4 MVP Integration	Nov 2 – Nov 15 Target: End W4	Integrated CV + ML + rep counting + feedback engine, with the UI showing exercise, reps, form status, and feedback.	Live session produces reps + feedback end to end.
M5 Advanced Features	Nov 9 – Nov 22 Target: End W5	Privacy layer (blurring), NLP/LLM assistant, model improvements, privacy and assistant UI.	Contextual answers + privacy demo.
M6 Deployment & Validation	Nov 16 – Nov 29 Target: End W6	Dockerized application, Azure deployment, MLflow configuration, final evaluation (F1-score, confusion matrix), FPS/latency benchmarking.	Deployed system with KPI results reported.
M7 Finalization	Nov 23 – Dec 6 Target: End W7	Final integration, release version, documentation, presentation, demo environment.	Integrated demonstrable system, ready for submission on Dec 7.

1.6 Resource Allocation

Resource	Planned Use	Owner	Priority
Mendeley fitness video dataset	Primary MVP data source; Good/Bad labels and multiple views [1]	Technical Lead — Mahmoud; ML Engineer — Mohamed	High
Python + OpenCV	Video/frame processing and camera pipeline	CV Engineer — Mostafa; Backend — Hashem	High
MediaPipe Pose / optional YOLO Pose	Human pose/keypoint extraction	CV Engineer — Mostafa	High
Scikit-learn / XGBoost	Form classification experiments	ML Engineer — Mohamed	High
FastAPI + React/Streamlit	Backend API and UI	Backend — Hashem; Frontend — Islam	High
MLflow + Git/GitHub + Docker	Tracking, versioning, packaging	MLOps — Islam	High
Microsoft Azure	Cloud deployment	MLOps — Islam; Backend — Hashem	High
LLM API	Contextual fitness assistant	Backend — Hashem; Technical Lead — Mahmoud	Medium

1.7 Task Assignment & Roles

The team has five members. Each member holds one primary role, and **every member works on every milestone** through a role-specific task. By the end of the project, each member has contributed to all seven milestones, so the whole team gains experience across the full system.

1.7.1 Mahmoud Mosad Eraky

Milestone	Mahmoud's Task
M1 — Planning & Requirements	Define system architecture, AI pipeline, project scope, requirements and success criteria
M2 — Data & Preprocessing	Analyze dataset structure, define labels/classes, design preprocessing and feature schema
M3 — Pose & AI Baseline	Implement/experiment with pose-based feature extraction and establish first AI baseline
M4 — MVP Integration	Integrate CV + ML + Rep Counting and coordinate end-to-end testing
M5 — Advanced Features	Design how Privacy + NLP/LLM fit into the main architecture
M6 — Deployment & Validation	Analyze system performance, model results and deployment readiness
M7 — Finalization	Final technical review, documentation, presentation and demo coordination

1.7.2 Mostafa Darwish Mostafa

Milestone	Mostafa's Task
M1 — Planning & Requirements	Define CV requirements, camera assumptions, supported views and pose-estimation constraints
M2 — Data & Preprocessing	Prepare videos/frames, inspect image quality, preprocessing and pose-data generation
M3 — Pose & AI Baseline	Implement MediaPipe Pose / evaluate YOLO Pose and validate keypoints
M4 — MVP Integration	Integrate pose estimation with joint-angle calculations and real-time pipeline
M5 — Advanced Features	Implement privacy layer: face/body blurring and investigate robustness under occlusion
M6 — Deployment & Validation	Test FPS, pose latency, camera conditions and CV failure cases
M7 — Finalization	Document CV methodology and help prepare technical demonstration

1.7.3 Mohamed Tarek Mohamed

Milestone	Mohamed's Task
M1 — Planning & Requirements	Define ML problem formulation, classes, labels and evaluation metrics
M2 — Data & Preprocessing	Clean dataset, analyze class balance, create train/validation/test split and feature dataset
M3 — Pose & AI Baseline	Train baseline models: Random Forest, SVM, XGBoost and compare performance
M4 — MVP Integration	Integrate selected model into real-time inference and analyze misclassifications
M5 — Advanced Features	Experiment with model improvements, calibration and potentially temporal features
M6 — Deployment & Validation	Perform final evaluation, confusion matrix, F1-score, error analysis and performance benchmarking
M7 — Finalization	Prepare ML results, graphs, tables and technical explanation for presentation

1.7.4 Hashem Salah El-Din El-Husseini

Milestone	Hashem's Task
M1 — Planning & Requirements	Define backend requirements, APIs and real-time data flow
M2 — Data & Preprocessing	Build scripts/interfaces needed to feed processed pose/features into the application
M3 — Pose & AI Baseline	Build initial real-time inference pipeline connecting camera → CV → ML
M4 — MVP Integration	Implement repetition-counting state machine and feedback engine
M5 — Advanced Features	Integrate NLP + LLM API so users can ask questions about their exercise/form
M6 — Deployment & Validation	API testing, latency testing, error handling and backend optimization
M7 — Finalization	Final API integration, demo workflow and documentation

1.7.5 Islam Ahmed Hosny

Milestone	Islam's Task
M1 — Planning & Requirements	Define UI requirements, user workflow and deployment requirements
M2 — Data & Preprocessing	Set up Git/GitHub workflow, project environment and MLflow experiment structure
M3 — Pose & AI Baseline	Build initial UI/dashboard and connect it to the AI pipeline
M4 — MVP Integration	Display exercise, reps, form status, mistakes and feedback in the interface
M5 — Advanced Features	Integrate privacy controls and LLM assistant UI into the application
M6 — Deployment & Validation	Dockerize application, deploy to Azure, configure MLflow and monitor performance
M7 — Finalization	Final deployment, release version, demo environment and presentation visuals

1.8 Risk Assessment & Mitigation Plan

Risk	Probability	Impact	Mitigation	Contingency
Compressed 7-week schedule delays the critical path	High	High	Parallel tracks for all members, mock interfaces in W1, weekly gate reviews, MVP first, feature freeze at the end of W6.	Reduce LLM and privacy modules to minimal versions; protect the core MVP.
Integration problems between parallel workstreams	Medium	High	Interface contracts agreed in W1, mock endpoints, weekly integrated build, peer review.	Dedicated integration session between the affected members within 24 hours.
Dataset imbalance/insufficient examples	Medium	High	Inspect distribution early; use weighting/augmentation where appropriate; limit scope.	Use simpler baseline and report limitations.
Pose failures from occlusion/camera placement	High	High	Camera-position guidance, confidence checks, smoothing, supported views.	Prompt user to reposition camera.

Risk	Probability	Impact	Mitigation	Contingency
Subject overfitting	Medium	High	Subject-independent split.	Reduce model complexity and strengthen validation.
False/double rep counts	Medium	High	Two thresholds, hysteresis/state machine, smoothing, cooldown.	Use stable rule-based counter.
High real-time latency	Medium	Medium	Profile stages; reduce resolution/FPS; lightweight pose model.	Controlled FPS and required landmarks only.
Poor/unsafe LLM response	Medium	Medium	Structured context, constrained prompts, fallback; no medical diagnosis.	Treat LLM as explanatory assistant.
Azure integration delay	Medium	High	Set up the environment and MLflow in M2, integrate the MVP early in M4, and start Dockerization as soon as the MVP is stable.	Demonstrate local Docker deployment if necessary.
Team schedule conflicts or member unavailability	Medium	Medium	Weekly sprint planning, clear ownership, shared knowledge across tracks.	Reassign tasks to another member and prioritize MVP.
Privacy concerns	Medium	High	Minimize raw-video storage; anonymize identifiable regions.	Local processing and no persistent raw video.

1.9 Key Performance Indicators (KPIs)

KPI	Target / Measure	Purpose	Measurement
Form classification F1-score	Target ≥ 0.90 on held-out subjects (target, not a claimed result)	Balances false positives/negatives	Classification report
Exercise recognition accuracy	Target ≥ 0.90	Confirms exercise separation	Per-exercise accuracy
Rep-counting error	Preferably ≤ 1 rep per short test sequence; report MAE/count error	Practical counter quality	Manual vs predicted count
Real-time performance	Target ≥ 15 FPS end-to-end	Usable live interaction	FPS/latency profiling
Feedback latency	Target < 500 ms	Immediate feedback	Event timestamp measurement
Demo reliability	≥ 95% successful sessions	System stability	Session log
LLM relevance	≥ 90% on predefined question set	Assistant usefulness	Human evaluation
MLOps reproducibility	100% of final experiments traceable	Engineering quality	MLflow + Git audit
On-time delivery	All 7 milestones passed at gate review; final delivery by Dec 7, 2026	Schedule control	Weekly gate review log

2. Literature Review

The review examines established pose-estimation technology and comparable fitness-form systems to identify useful design patterns, limitations, and opportunities for differentiation. Academic sources provide the technical foundation, while open-source systems provide practical implementation references.

2.1 Review Approach

- Review academic sources for real-time pose-estimation foundations and perception pipelines.
- Review open-source fitness systems for rep counting, joint-angle analysis, form feedback, and user interfaces.
- Review an existing labeled fitness video dataset to validate the proposed data strategy.
- Compare solutions using exercise coverage, form assessment, rep counting, privacy, conversational AI, MLOps, and deployment.

2.2 Similar Solutions and Projects

Solution	Capabilities	Strengths	Limitations	How We Use the Insight
BlazePose / MediaPipe Pose [2]	33-keypoint real-time pose estimation.	Lightweight, real-time, fitness-oriented.	Pose layer only; not a complete fitness assistant.	Use as core pose-estimation layer.
Workout Form Detector [3]	Biceps Curl, Squat, Push-up; rep counting; form feedback; webcam/video.	Very close to MVP; practical stage tracking.	Primarily rule/threshold based; limited MLOps/LLM/privacy.	Reference rep/form architecture; implement our own logic.
SmartForm [4]	On-device curls, squats, push-ups; calibration; form-gated reps; rep quality; summaries.	Strong privacy/on-device design and advanced rep logic.	Mobile-specific and broader than MVP.	Inspire calibration and rep-quality ideas.
FitVision-AI [5]	Pose tracking, rep counting, form correction, voice guidance, workout logging.	Broad user-facing functionality; same three core exercises.	Mostly rule-based; not centered on MLflow/LLM/cloud.	Reference UI, feedback, and session logging.
FitVisionAI [6]	MediaPipe, joint angles, ANN exercise classification, rep counting, form feedback.	Shows pose + learned classifier integration.	Less emphasis on privacy, LLM, MLOps.	Supports our hybrid pose-feature + ML approach.

2.3 Comparative Analysis

Capability	Our Project	Workout Form Detector	SmartForm	FitVision-AI	FitVisionAI	Decision
Pose estimation	✓	✓	✓	✓	✓	MediaPipe Pose for MVP
3 MVP exercises	✓	✓	✓	✓	Partial	Keep focused scope
ML form classifier	✓	Limited	Limited	Rule-based	✓ ANN	Compare RF/SVM/XGBoost
Rep counting	✓	✓	✓	✓	✓	State machine + thresholds
Real-time feedback	✓	✓	✓	✓	✓	Required
Privacy / blur	✓	—	On-device privacy	—	—	Add privacy layer
NLP / LLM assistant	✓	—	—	—	—	Differentiating feature
MLOps / MLflow	✓	—	—	—	—	Engineering differentiator
Azure deployment	✓	—	—	—	—	Cloud target

> **Literature Review Conclusion**
> Existing systems show that real-time pose estimation, joint-angle analysis, repetition counting, and form feedback are feasible. Our proposed system differentiates itself by combining learned form classification with privacy-aware processing, an NLP/LLM assistant, MLOps practices, and Azure deployment.

2.4 Lecturer Feedback & Evaluation

The lecturer reviewed the project work, including the proposal, scope, plan, literature review, and requirements. The overall feedback was positive: the work is on track, the approach is sound, and the project can proceed as planned. No major changes were requested.

Area	Evidence Reviewed	Lecturer Feedback	Status
Problem & scope	Proposal + scope	Scope is clear and achievable.	Approved
Technical approach	Architecture + stack	Pipeline and technology choices are appropriate.	Approved
Data strategy	Dataset + preprocessing	Data source and preprocessing plan are adequate.	Approved
Evaluation	KPIs + tests	Metrics and targets are realistic.	Approved
Innovation	Privacy + LLM + MLOps + Azure	Advanced components add clear value.	Approved
Presentation	Demo plan	Demo plan is clear.	Approved

> **Lecturer Review Outcome**
> The project plan and documentation were reviewed and accepted. The team will continue with the plan as documented.

2.5 Suggested Improvements

- Finish a stable MVP before adding temporal deep learning.
- Define interface contracts in Week 1 so that all five members can work in parallel from the start.
- Use subject-independent evaluation to measure generalization.
- Add calibration/adaptive thresholds to reduce sensitivity to body proportions and camera distance.
- Send structured exercise-analysis context to the LLM instead of unrestricted raw video.
- Track experiments in MLflow and keep preprocessing configurations reproducible.
- Profile real-time performance early to avoid deployment bottlenecks.
- Treat privacy as a system requirement: minimize raw-video retention and anonymize visual data when retained/displayed.

2.6 Proposed Final Grading Criteria

This is an internal proposed rubric aligned with the assignment wording. It is not presented as the lecturer's official grading policy and should be updated if an official rubric is supplied.

Category	Suggested Weight	Evaluation	Evidence
Documentation & Planning	20%	Planning, scope, literature review, requirements, risks, architecture	Documentation
Implementation	40%	Pose pipeline, form analysis, rep counting, feedback, integration, advanced components	Prototype + code
Testing & Evaluation	20%	Test cases, metrics, error analysis, performance, reliability	Test report
Presentation & Demonstration	20%	Technical explanation, demo, team participation, Q&A	Presentation + demo

3. Requirements Gathering

Requirements were derived from the intended users, the defined MVP, the technical architecture, and the need to demonstrate an end-to-end AI engineering workflow within the seven-week schedule.

3.1 Stakeholder Analysis

Stakeholder	Role / Interest	Key Needs	Priority	Success Signal
End User / Trainee	Uses system during exercise	Accurate feedback, rep count, low latency, simple UI, privacy	High	Completes a session and understands feedback
Project Team (5 members)	Builds/maintains system	Clear scope, reproducible pipeline, modular code, KPIs	High	All modules integrate
Lecturer / Evaluator	Reviews technical/project quality	Clear requirements, justified methods, credible evaluation	High	Meets academic criteria
Fitness/Domain Reviewer	Validates exercise logic	Meaningful form criteria and feedback	Medium	Feedback is reasonable
Cloud/Platform Operator	Supports deployment	Reliable deployment, logs, efficiency, security	Medium	Stable deployed service

3.2 User Stories

ID	User Story	Priority	Acceptance Criteria
US-01	As a trainee, I want to select an exercise so the correct analysis is applied.	High	Squat, Push-up, or Biceps Curl can be selected.
US-02	As a trainee, I want the camera to detect my body pose automatically.	High	Body landmarks are detected with sufficient confidence.
US-03	As a trainee, I want repetitions counted automatically.	High	Each completed cycle increments once.
US-04	As a trainee, I want incorrect form identified so I can correct it.	High	Status and relevant issue are displayed.
US-05	As a trainee, I want immediate feedback during movement.	High	Feedback meets latency target.
US-06	As a privacy-conscious user, I want identifiable regions blurred when needed.	Medium	Privacy mode applies configured anonymization.
US-07	As a trainee, I want to ask the AI assistant questions about my detected mistakes.	Medium	LLM answers using current exercise context when available.
US-08	As a team member, I want experiments tracked for reproducibility.	High	Final experiments have MLflow records.
US-09	As an evaluator, I want objective performance metrics.	High	Report includes classification and rep-counting metrics.

3.3 Use Cases

Use Case	Actor	Precondition	Main Flow	Expected Result
UC-01 Start Workout	Trainee	Camera available	Select exercise → enable camera → pose detected → start	Live analysis begins.
UC-02 Count Repetitions	Trainee	Pose stable	Compute angle → update state → detect completion → increment	Rep count updates once.
UC-03 Evaluate Form	Trainee	Features available	Extract features → model/rules → classify → feedback	Form status + feedback displayed.
UC-04 Ask AI Assistant	Trainee	LLM service available	Enter question → add context → call LLM → display answer	Contextual explanation returned.
UC-05 Enable Privacy Mode	Trainee	Camera active	Enable privacy → detect region → blur → continue	Analysis continues with reduced visual exposure.
UC-06 Track Experiment	Team Member	Pipeline configured	Run → log parameters/metrics → artifact	Experiment traceable.
UC-07 Deploy System	Team Member	Docker image available	Build → deploy → configure → health check	System available in target Azure environment.

3.4 Functional Requirements

ID	Requirement	Priority	Verification
FR-01	Capture frames from supported camera/video input.	High	Functional test
FR-02	Detect human pose landmarks for one user.	High	Pose test
FR-03	Support Squat, Push-up, Biceps Curl.	High	Exercise test
FR-04	Calculate exercise-specific joint angles and movement features.	High	Feature validation
FR-05	Classify form as correct/incorrect.	High	Model evaluation
FR-06	Count completed repetitions using state/threshold logic.	High	Rep-count test
FR-07	Display exercise, reps, form status, and feedback.	High	UI test
FR-08	Handle low-confidence pose detections without misleading counts where possible.	High	Edge-case test
FR-09	Provide privacy/anonymization mode.	Medium	Privacy test
FR-10	Provide NLP/LLM assistant for exercise-analysis questions.	Medium	QA test
FR-11	Track experiments/model versions with MLflow.	High	MLOps audit
FR-12	Package application using Docker.	High	Deployment test
FR-13	Deploy integrated system to Microsoft Azure.	High	Cloud test
FR-14	Record relevant session metrics without permanent raw-video storage.	Medium	Data-handling test

3.5 Non-functional Requirements

Category	Requirement	Target / Criterion	Verification
Performance	Real-time processing	Target ≥ 15 FPS end-to-end; report actual result	Profiling
Latency	Feedback responsiveness	Target < 500 ms	Timestamped measurement
Accuracy	Form classification	Target F1 ≥ 0.90 on held-out subjects; report actual result	Evaluation
Reliability	Session stability	≥ 95% successful test/demo sessions	Session log
Usability	Ease of use	First-time user can start a supported exercise without technical help	Usability test
Security	Secrets/API protection	Environment variables; secured endpoints as needed	Security checklist
Privacy	Data minimization	No unnecessary permanent raw-video storage; anonymize when needed	Privacy review
Maintainability	Modular architecture	Separate CV, ML, backend, UI, MLOps modules	Code review
Reproducibility	Experiment traceability	Final model linked to code/config/data version	MLflow audit
Scalability	Deployment	Architecture allows scaling if required	Architecture review
Availability	Service stability	Suitable for development/demo deployment	Health check
Safety	No medical diagnosis	Feedback is technique guidance, not clinical advice	Content review

3.6 Assumptions and Constraints

- **The project must be delivered by Monday, December 7, 2026**, giving a seven-week working period (October 19 – December 6, 2026).
- All five members work in parallel on every milestone; each member has a defined primary role and contributes to every milestone.
- The MVP is limited to one person at a time.
- The main data source is an existing labeled fitness video dataset; no large new data-collection campaign is planned because of schedule constraints.
- The Mendeley dataset provides Good/Bad labels, exercise classes, subject identity, and multiple views [1].
- Pose quality depends on visibility, camera placement, lighting, occlusion, and body position.
- Rep counting is initially rule/state-machine based rather than dependent on a dedicated supervised repetition-count dataset.
- LLM functionality depends on external API availability and is not a medical/clinical advisor.
- Azure deployment is performed in M6, after the integrated MVP is stable in M4, with a local Docker deployment as fallback.
- No new features are added after the feature freeze at the end of Week 6 (November 29, 2026).

Requirements Traceability Matrix

Objective	Requirements	KPI	Test Evidence	Milestone
Form analysis	FR-04, FR-05	F1-score	Classification report + confusion matrix	M3 (End W3); M6 (End W6)
Rep counting	FR-06	Rep-count error	Manual vs predicted count	M4 (End W4)
Real-time feedback	FR-07, FR-08	FPS + latency	Performance test	M4 (End W4)
Privacy	FR-09, FR-14	Privacy checklist	Privacy test	M5 (End W5)
AI assistant	FR-10	Answer relevance	Human evaluation	M5 (End W5)
MLOps	FR-11, FR-12	Traceability rate	MLflow/Git/Docker audit	M2 (End W2); M6 (End W6)
Cloud deployment	FR-13	Deployment/health checks	Azure evidence	M6 (End W6)

References

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